MATH 526 Section 001 Lab 12 Spring 2006

Part I: Eigenvalues and Eigenvectors

A number λ is an eigenvalue of a square matrix $\mathbf{A}$ if and only if there is a vector $\mathbf{v} \neq \mathbf{0}$ such that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. MATLAB has a built-in command **eig** for the computation of eigenvalues and eigenvectors.

Create the matrix

$$\mathbf{A} = \begin{bmatrix} -1.5 & -1.5 & -1 \\ 2 & 2 & 1 \\ -1 & -1 & -0.5 \end{bmatrix}$$

>>**eig(A)**

This returns a vector containing the eigenvalues of **A**.

>>**[V, D] = eig(A)**

This returns a diagonal matrix **D** whose diagonal components are the eigenvalues of **A**, and a matrix **V** whose columns are the corresponding eigenvectors of **A**.

>>**v = V(:,1)**

>>**A*v**

>>**D(1,1)*v**

This checks that the first column of **V** is an eigenvector of **A** corresponding to the first eigenvalue of **A** on the diagonal of **D**.

A generic $n \times n$ matrix has n distinct eigenvalues. This can be seen by generating a 3×3 matrix randomly and computing its eigenvalues.

>>**C = rand(3)**

This creates a 3×3 matrix randomly by the MATLAB random number generator.

>>**eig(C)**

The eigenvalues of **C** should be distinct unless this is your lucky day. You can repeat this many times in various dimensions, and the eigenvalues should be distinct almost all the time.

Clear all variables before you work on Part II.